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Motivational Interviewing Chatbot for Alcohol-Related Dialogues

A Fine-Tuned, Retrieval-Augmented, Safety-Layered LLM Pipeline

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Abstract

High-risk alcohol use is among the most preventable health burdens worldwide, yet the evidence-based behavioral approach to addressing it – Motivational Interviewing (MI) – requires a trained counselor and does not scale. General-purpose large language models (LLMs) appear at first to be a natural fit, but they default to advice-giving and gentle confrontation, which is precisely what drives ambivalent or defensive users away.

This capstone project designs, implements, and evaluates an end-to-end MI chatbot system that addresses this gap. The contribution is not a single model but a six-stage pipeline: (1) input safety screening for crisis ideation and restricted topics, (2) silent state inference scoring the user across seven emotion classes and four defensiveness levels, (3) retrieval-augmented generation over a curated MI knowledge base using FAISS and `all-MiniLM-L6-v2` embeddings, (4) response generation by a LoRA-fine-tuned `gemma-2-2b-it` served locally through Ollama, (5) output safety screening for medical-advice patterns, and (6) per-turn JSONL audit logging.

The fine-tuning corpus combines the alcohol-only subset of AnnoMI [1] with approximately one thousand synthetic MI conversations generated by Claude under explicit MI guardrails. The system is evaluated using an LLM-as-judge framework [2] in which Claude Sonnet scores responses on a 1–5 scale across five MI dimensions: empathy, MI technique, resistance handling, autonomy support, and appropriateness. Across 181 evaluation items spanning real AnnoMI turns, synthetic scenarios, and edge cases, the fine-tuned model with RAG (*v2 + RAG*) reaches an overall score of 3.81/5.0, a relative improvement of +17.2% over the base `gemma-2-2b-it` (3.25/5.0) and closing roughly 40% of the gap to a much larger Claude Haiku reference (4.61/5.0). The system is deployed via two interfaces – a Streamlit developer/evaluation app and a React + FastAPI end-user web app – both sharing a single Python pipeline. All conversations are auditable through structured per-turn JSONL logs.

Keywords: Motivational Interviewing, large language models, parameter-efficient fine-tuning, LoRA, retrieval-augmented generation, LLM-as-judge evaluation, conversational safety, behavioral health.

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1 Introduction and Background

1.1 Problem Statement

Roughly one in seven adults in the United States meets criteria for an alcohol-use disorder in any given year, yet only a small minority ever speak to a counselor about it [3]. The evidence-based clinical approach to ambivalence about substance use is *Motivational Interviewing* (MI), developed by Miller and Rollnick [4]. MI is collaborative and goal-oriented but explicitly non-confrontational: rather than tell the patient what to do, the clinician helps the patient articulate their own reasons for change.

Two facts motivate this project. First, qualified MI clinicians do not scale – training, supervision, and licensure are slow and expensive, and access is geographically uneven. Second, off-the-shelf LLMs do not, by default, behave like MI counselors. They are trained on instruction-following corpora that reward direct answers and informative explanations, so when faced with a user who says “*I don’t think I drink that much*”, the modal response from a generic assistant is to either reassure the user (a missed opportunity to develop discrepancy) or to enumerate health risks (a frontal assault that provokes resistance). Neither is MI.

The question this capstone addresses is therefore practical:

Can a small, locally-deployable language model be turned into a credible Motivational Interviewing assistant for alcohol-related conversations, with explicit safety boundaries and full conversation auditability?

1.2 Project Goals and Objectives

The objectives of this project, in priority order, are:

1. **Behavioral fidelity.** Produce responses that visibly follow MI principles (express empathy, roll with resistance, support autonomy, develop discrepancy) rather than defaulting to advice-giving or information-dumping.
2. **State awareness.** Detect when a user shifts from openness to defensiveness mid-conversation and adapt the response style accordingly, silently – without exposing the inferred state to the user.
3. **Safety.** Detect crisis ideation (suicidal language, self-harm, overdose) and restricted-topic content (medication dosing, prescriptions) both before *and* after model inference, and short-circuit to evidence-based crisis resources when warranted.
4. **Auditability.** Produce a structured, per-turn record of every conversation containing the inferred state, retrieved knowledge, safety flags, and full text – so that the pipeline is inspectable rather than a black-box LLM call.
5. **Local deployability.** Run the production model on a CPU-only workstation through Ollama, with a transparent fallback to a hosted API when local inference is unavailable.

1.3 Scope and Limits

The system is a research artifact, not a clinical product. It is not validated under an institutional review board, has not been reviewed at scale by licensed clinicians, and is not appropriate for unsupervised deployment to people in acute distress. Its purpose is to demonstrate the engineering of an MI-aligned conversational pipeline and to characterize, quantitatively, what a small fine-tuned model can reach versus a large reference model on the same task. The honest limits of the system are stated explicitly in Section 8.

1.4 Report Outline

The remainder of this report is organized as follows. Section 2 surveys related work in MI, parameter-efficient fine-tuning, retrieval-augmented generation, and LLM-as-judge evaluation. Section 3 describes the six-stage pipeline architecture and the fine-tuning, retrieval, and safety designs. Section 4 discusses the key engineering analyses and trade-offs – data scarcity, model size selection, evaluation design. Section 5 specifies the evaluation set, the LLM-as-judge protocol, and the configurations compared. Section 6 presents quantitative and qualitative results. Section 7 summarizes findings, and Section 8 lays out recommended future work.

2 Survey of Related Literature

This project sits at the intersection of four bodies of work: Motivational Interviewing as a behavioral-health methodology, parameter-efficient fine-tuning of language models, retrieval-augmented generation, and automated evaluation of dialogue systems. The relevant prior art in each is summarized here.

2.1 Motivational Interviewing

Motivational Interviewing was introduced by Miller and Rollnick [4] as a counseling style for resolving ambivalence about behavior change. Its operational core is the OARS skill set – *Open questions*, *Affirmations*, *Reflections*, *Summaries* – combined with four guiding principles: *express empathy*, *develop discrepancy*, *roll with resistance*, and *support self-efficacy*. Meta-analytic evidence supports MI as an effective intervention for alcohol-use disorders compared to no-treatment controls, with effect sizes that, while modest, are robust across populations and settings [5].

The empirical literature on *automating* MI-aligned dialogue is much younger. Wu et al. [1] introduced the AnnoMI corpus – the only publicly available dataset of expert-annotated MI transcripts – and a baseline classifier for MI-consistent versus MI-inconsistent turns. AnnoMI is the real-data backbone of this project.

2.2 Parameter-Efficient Fine-Tuning

Full fine-tuning of multi-billion-parameter language models is prohibitive on single-GPU consumer hardware. Hu et al. [6] introduced **Low-Rank Adaptation (LoRA)**, which freezes the pretrained weight matrix $W \in \mathbb{R}^{d \times k}$ and learns a low-rank update $\Delta W = BA$ where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$, with $r \ll \min(d, k)$. LoRA reduces trainable parameters by orders of magnitude while matching full-fine-tuning quality on GLUE-scale benchmarks. Dettmers et al. [7] extended this to QLoRA, in which the frozen base model is quantized to 4-bit precision so that even very large models fit on a single consumer GPU.

This project uses LoRA via the Hugging Face PEFT library [8], applied to all attention and MLP projection matrices of `gemma-2-2b-it`, with the base weights kept in 4-bit NF4 precision during training. The Gemma model family itself was released by Google [9], with the 2B-instruct checkpoint chosen here for local deployability.

2.3 Retrieval-Augmented Generation

Lewis et al. [10] introduced **Retrieval-Augmented Generation (RAG)** as a method for grounding language model outputs in non-parametric knowledge: at inference time, an embedding-based retriever fetches the k most relevant passages from a document store and concatenates them to the prompt. RAG mitigates hallucination and lets the system update its knowledge without retraining the model.

The retrieval component of this project uses FAISS [11] as the approximate-nearest-neighbor index and `all-MiniLM-L6-v2` [12] as the encoder. `all-MiniLM-L6-v2` was selected over larger encoders for its $\approx 30\times$ speed advantage on CPU and its strong retrieval quality on the MTEB benchmark [13] relative to its size.

2.4 LLM-as-Judge Evaluation

Evaluating open-ended dialogue is a long-standing problem. Reference-based metrics such as BLEU and ROUGE correlate poorly with human judgement on tasks where many distinct responses are valid [14]. Human evaluation is the gold standard but is too slow to drive an iteration loop on a one-person project.

Zheng et al. [2] demonstrated that strong LLMs (GPT-4, Claude) used as judges achieve 80%+ agreement with human raters on chatbot-arena-style pairwise judgements, comparable to inter-human agreement. Subsequent work [15] showed that the LLM-as-judge protocol is sensitive to position bias, verbosity bias, and self-preference bias, and recommends mitigations including symmetric pairwise prompting and explicit rubrics. This project applies the rubric variant – the judge sees a fixed five-dimension MI rubric and produces calibrated 1–5 scores – and treats the results as a coarse iteration signal rather than ground truth, in line with the recommendations of [15].

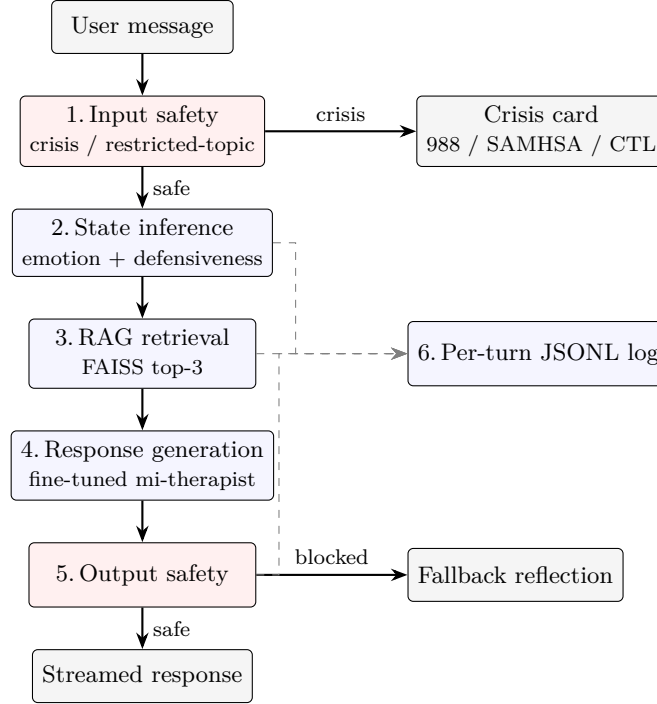


Figure 1: Six-stage runtime pipeline. Solid arrows are the request path; dashed arrows indicate side-effect writes to the per-turn audit log. Stages 1 and 5 are red because they can short-circuit the pipeline; the remaining stages always execute when the input passes the input safety check.

2.5 Conversational Safety

Safety in deployed conversational AI is typically implemented as a defense-in-depth stack: input filtering, in-context guardrails inside the prompt, post-generation classifiers, and audit logging [16, 17]. This project adopts a stripped-down version of that pattern – a regex-based input and output screen for crisis and medical-boundary content, plus per-turn JSONL audit logging – chosen explicitly because it is auditable line-by-line, which an embedding-based classifier is not.

3 Methodology

3.1 System Overview

The system is a six-stage Python pipeline shared by two front ends. Every user message is run through the same pipeline; only the presentation layer differs between the developer/evaluation interface and the end-user web interface. Figure 1 shows the runtime path of a single turn.

The orchestrator is `ConversationManager`, which owns a `ConversationHistory`, a `StateInferencer`, a `RAGPipeline`, a `ResponseGenerator`, and a `ConversationLogger`. The same object graph is wrapped by either the Streamlit dev/eval app (`app.py`) or the FastAPI server (`backend_server.py`).

3.2 Stage 1 – Input Safety

The input safety layer (`src/safety_layer.py`) screens the incoming user message for two categories of content:

- **Crisis indicators.** Suicidal ideation (active and passive), self-harm intent, and overdose. Patterns include direct phrases (“kill myself”, “end it”) and indirect phrasings (“no reason to keep going”, “don’t even want to be here”). The original implementation used strict-adjacency regexes; these missed many natural phrasings, so the patterns were broadened with optional-intensifier groups during late-stage polish.
- **Restricted topics.** Medication dosing and prescription advice. The system is not a clinical authority; it is explicitly designed to refuse this category of request.

A crisis match short-circuits the entire pipeline – no RAG retrieval, no LLM call, just a pre-written response containing the 988 Suicide & Crisis Lifeline, SAMHSA helpline, and Crisis Text Line. A restricted-topic match returns a medical-boundary response with the same short-circuit.

The choice of regex over an embedding-based classifier is deliberate: every trigger is auditable line-by-line, and the cost of a false negative here is much higher than the cost of a false positive.

3.3 Stage 2 – State Inference

The state inferencer (`src/state_inference.py`) scores each user message on two axes:

- **Emotion** – one of seven classes: *neutral, frustrated, anxious, sad, angry, hopeful, contemplative*.
- **Defensiveness** – one of four levels: *none, low, moderate, high*.

For local Gemma runs the inference is rule-based using weighted regular expressions over linguistic markers (modal verbs, hedging, negations, intensity words), which runs in ~30 ms. For API runs an LLM call is used instead. Either path produces the same JSON structure, which is consumed downstream.

The inferred state *never appears in the response surface*. It is used silently to weight retrieval and to select among prompt-template variants. This is a deliberate UX decision – being told “I detect that you are defensive” would itself be MI-inconsistent.

3.4 Stage 3 – Retrieval-Augmented Generation

The retrieval component (`src/rag_pipeline.py`) maintains a FAISS index over a curated MI knowledge base. The corpus consists of nine markdown documents authored for this project plus an extracted set of high-quality MI exchanges from AnnoMI (Table 1).

The configuration is:

- **Encoder.** `sentence-transformers/all-MiniLM-L6-v2` (384-dim, English-only).

Table 1: Contents of the MI knowledge base used for retrieval.

Document	Purpose
mi_principles.md	Four guiding MI principles
oars_detailed_guide.md	OARS skill set with examples
mi_antipatterns.md	What NOT to say
handling_resistance.md	Roll-with-resistance strategies
defensiveness_playbook.md	Tactics by defensiveness level
stage_of_change_strategies.md	Stage-matched interventions
emotional_responses.md	Reflection patterns by emotion
harm_reduction.md	Harm-reduction framings
crisis_protocols.md	Escalation and resource scripts
annomi_examples/	Curated AnnoMI excerpts

- **Chunking.** `chunk_size` = 500 characters, `overlap` = 50.
- **Index.** FAISS `IndexFlatL2` (exact, sufficient given the small corpus size).
- **Retrieval.** Top- k with $k = 3$, biased by inferred state – e.g., a turn scored as “high” defensiveness oversamples the `handling_resistance.md` and `defensiveness_playbook.md` chunks instead of empathy reflections.

The retrieved chunks are injected into the prompt as “context-specific guidance” in stage 4.

3.5 Stage 4 – Response Generation

The response generator (`src/response_generator.py`) builds the final prompt as the concatenation of:

1. a system message describing MI principles and style guidelines,
2. the top-3 retrieved knowledge chunks,
3. a sliding context window of recent dialogue turns,
4. the new user message.

The prompt is sent to the fine-tuned `mi-therapist` model running on Ollama. If Ollama is unreachable, the system falls back transparently to Claude Haiku via the Anthropic API, with the fallback fact recorded in the audit log.

Fine-Tuning Configuration

The trained adapters are merged back into the base weights and exported to 4-bit GGUF (Q4_K_M) for inference, producing a ≈ 1.6 GB artifact that runs on a CPU through Ollama with ~ 2 – 4 s per-turn latency.

Table 2: Fine-tuning configuration. “Effective batch size” is $1 \times 8 = 8$; “trainable params” figure depends on rank and target modules.

Parameter	Value
Base model	google/gemma-2-2b-it
Quantization (training)	4-bit NF4 (QLoRA-style)
Method	LoRA via PEFT
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
Target modules	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Optimizer	paged_adamw_8bit
Learning rate	2×10^{-4}
Scheduler	cosine with 10% warmup
Epochs	3
Batch size	1 (per device)
Gradient accumulation	8
Precision	bf16
Gradient checkpointing	enabled
Max sequence length	2048 tokens
Hardware	Google Colab T4 (16 GB)

3.6 Stage 5 – Output Safety

After generation, the response is screened a second time for medical-advice patterns (“you should take”, “I recommend stopping”, explicit dosage references). If matched, the LLM output is replaced with a generic reflective fallback. This stage exists because system-prompt guardrails alone are not reliable – a fine-tuned model can still drift, especially on adversarial inputs.

3.7 Stage 6 – Audit Logging

The `ConversationLogger` writes one JSONL record per turn under `logs/<session_id>.jsonl` containing: timestamp, user message, bot response, inferred state (emotion, defensiveness, reasoning, linguistic markers), safety level, retrieved chunks. Crisis-flagged turns are additionally appended to `logs/flagged_for_review.jsonl` for manual review. The audit log is the artifact that makes the pipeline inspectable rather than a black box.

3.8 Two Interfaces, One Pipeline

The same `ConversationManager` powers both UIs. Table 3 contrasts them.

The split exists because the visibility a developer needs (raw inferred state, retrieved chunks, fallback indicator) is the same visibility an end user should *not* see – it would break the conversational illusion that MI requires.

Table 3: The same backend pipeline is wrapped by two different front ends.

	End-user web app	Developer / eval app
Entry point	<code>backend_server.py</code> + React frontend	<code>app.py</code> (Streamlit)
Stack	FastAPI + React 18 (UMD) + Babel Standalone	Streamlit
Surfaces internal state?	No – clean conversation only	Yes – model picker, detected emotion, retrieval count, JSON debug panel
Crisis handling	Soft inline card with 988 / SAMHSA / CTL	Long-form text response
Audience	Non-technical demos, prof review	Engineering iteration and evaluation

3.9 Project Layout

Figure 2 maps the runtime pipeline to the source tree.

```

src/                                # Runtime pipeline (shared by both UIs)
  conversation_manager.py           # Orchestrator
  state_inference.py                # Stage 2: emotion + defensiveness
  rag_pipeline.py                   # Stage 3: FAISS-based retrieval
  response_generator.py             # Stage 4: adaptive prompt construction
  safety_layer.py                   # Stages 1 and 5 + Stage 6 logging
  config.py

backend_server.py                   # FastAPI server (end-user web app)
app.py                             # Streamlit dev/evaluation app

capstone-chatbot-design/           # End-user web frontend (React + Babel)
scripts/                           # Offline data + eval pipelines
notebooks/                         # Fine-tuning + inference notebooks
knowledge_base/                    # MI guidelines + AnnoMI excerpts
data/                              # Processed datasets, synthetic batches, eval results
models/                            # (gitignored) GGUF weights + Ollama Modelfile
logs/                              # (gitignored) per-session JSONL audit logs

```

Figure 2: Project layout. The `src/` package is the runtime, sharable between front ends; `scripts/` and `notebooks/` are offline.

4 Analysis

This section covers the engineering analyses that drove the major design choices: how to handle MI-data scarcity, how to choose a base model size, how to design an evaluation that actually informed iteration, and how to structure safety so it stayed auditable.

4.1 Data Scarcity and Synthetic Augmentation

The first practical constraint was data. AnnoMI [1], the only public corpus of expert-annotated MI dialogue, contains 133 transcripts across multiple behavioral-health domains. The alcohol-only filter reduces this to a small slice – after preprocessing into the project’s instruction-tuning format, only ≈ 363 training and ≈ 40 evaluation examples remain. That is not enough material on which to train a small language model to exhibit a consistent stylistic identity.

The choice was either to (a) train on the alcohol slice as-is and accept overfit-style brittleness, (b) widen the source to other AnnoMI domains and introduce off-topic drift, or (c) generate additional in-domain MI dialogue synthetically. Option (c) was selected. The synthetic generator (`scripts/generate_synthetic_data.py`) prompts Claude to simulate both sides of a counseling session, parameterized by scenario type:

- **short** – single-turn exchange,
- **medium** – 3–5 turns,
- **long** – 8–12 turns,
- **mixed defensiveness** – the user shifts between open and resistant within a single conversation.

The generator prompt embeds explicit MI guardrails: no advice-giving, no closed questions where open ones would do, the simple-then-complex reflection pattern, varied openers to prevent template collapse. Generation was run in batches of ≈ 50 –100 conversations and saved as JSONL under `data/synthetic/`, then deduplicated and merged with the AnnoMI subset.

The trade-off is honest: synthetic data is generated by a model that does not itself experience ambivalence, so the simulated client side is a stylized imitation rather than a faithful reproduction of human ambivalence. The evidence from evaluation (Section 6) is that despite this limitation, the augmented training set substantially improves response style on real AnnoMI test turns – so the synthetic dialogues are at least teaching the right output distribution.

4.2 Model Size and Local Deployability

A second decision was the base model. The constraint of running locally on a CPU through Ollama eliminated 7B-and-up models for interactive latency reasons. Among the small instruction-tuned families available at project start, three were candidates: `Llama-3.2-1B-Instruct`, `Phi-3-mini-4k-Instruct` (3.8B), and `Gemma-2-2b-it`. Gemma-2-2B was selected on three grounds:

1. **Quantization quality.** Gemma-2 quantizes to 4-bit GGUF with minimal observable degradation on the project’s eval set, an expected outcome given Gemma’s training-side quantization-friendliness.
2. **License.** Permissive enough for an academic capstone that redistributes a fine-tuned variant.

3. **Tooling.** First-class support in PEFT, in TRL’s SFTTrainer, and in Ollama as a target format.

The 2B size yielded ≈ 1.6 GB GGUF after Q4_K_M quantization, with $\sim 2\text{--}4$ s per-turn latency on a CPU. Larger models would close more of the quality gap to Claude (Section 6) but at a latency cost that breaks the conversational feel.

4.3 Evaluation Design

The single most important methodological decision was committing to LLM-as-judge evaluation rather than human ratings. Human MI scoring requires trained raters, $\approx 5\text{--}10$ minutes per dialogue for high-quality annotation, and inter-rater adjudication. None of that fits the iteration loop of a one-person semester project.

The judge is Claude Sonnet, prompted with a fixed five-dimension rubric (empathy, MI technique, resistance handling, autonomy support, appropriateness), each scored 1–5 with a written rationale. The rubric is reproduced in Section 5. Three design choices follow from the recommendations of Chiang and Lee [15] on LLM-judge bias:

- **Rubric-based, not pairwise.** Pairwise judgements are susceptible to position bias and verbosity bias; absolute scoring against a rubric is more robust at the cost of less calibrated individual scores. Since the goal here is comparing configurations *against the same rubric*, absolute scoring is sufficient.
- **Held-out judge.** The judge model (Claude Sonnet) is from a different provider than any system-under-test other than the Claude Haiku reference; this avoids the most pathological self-preference bias.
- **Coarse signal, not ground truth.** The scores are treated as an iteration signal. Configurations are not declared “better” on the basis of a 0.05 difference; only effects large enough to be visible to a human reader are emphasized in Section 6.

4.4 Safety as an Auditable Layer

The safety design uses regex rather than an embedding classifier. The argument for an embedding model is generalization – it would catch crisis ideation phrased in ways the regex misses. The argument against, in this setting, is auditability: a regex pattern list is line-by-line inspectable, and a regression in coverage is a one-line diff. An embedding-based classifier on the same problem requires either a labeled crisis corpus (which the project does not have) or an off-the-shelf classifier whose decision boundaries are not inspectable.

The cost of the regex choice was made visible mid-project: the original patterns required strict adjacency between intent words and self-references, so phrasings like “don’t even want to be here” were missed. The fix was adding optional-intensifier groups to the patterns – a one-commit change. That fix would not have been a one-commit change against an opaque classifier.

The defense in depth is the audit log. Even in cases where the regex misses a borderline phrasing, the per-turn JSONL record allows after-the-fact discovery of the miss, which an embedding classifier would not provide.

4.5 What This Analysis Yielded

The analyses above produced four concrete commitments that shaped the rest of the project:

- Synthetic data is part of training, not just real AnnoMI.
- The base model is Gemma-2-2B, not anything larger.
- The eval is LLM-as-judge with a rubric, run on a mixed test set spanning real AnnoMI turns, synthetic scenarios, and edge cases.
- Safety is a two-stage regex screen with audit logging, not an embedding classifier.

Each of these is revisited in Section 8 with the cases under which it should be reconsidered.

5 Experimental Setup

This section specifies the data, the evaluation protocol, and the configurations compared. The numerical results are deferred to Section 6.

5.1 Datasets

Training data

Training combines two sources, merged into a single instruction-tuned JSONL:

- **AnnoMI alcohol-only subset.** Filtered from the full AnnoMI corpus [1] to dialogues whose declared topic is alcohol use, converted into instruction/response pairs for SFT. After preprocessing, 363 training examples and 40 held-out evaluation examples remain.
- **Synthetic MI conversations.** Generated by the pipeline described in Section 4 and stored as 13 batch JSONL files under `data/synthetic/`, totalling approximately one thousand conversations across short, medium, long, and mixed-defensiveness scenario types.

Knowledge base for retrieval

The retrieval corpus is described in Table 1. It is held strictly separate from the training corpus, so retrieved content is not memorized text the model has already seen during fine-tuning.

Evaluation set

The evaluation set contains 181 user turns drawn from three sources:

- ~80 real AnnoMI turns (held out from training),
- ~80 synthetic eval turns spanning emotion $\times$ defensiveness combinations,
- ~20 hand-crafted edge cases (deeply ambivalent, sarcastic, minimizing, and crisis-adjacent phrasings).

Each item provides a user turn, a reference response (where available), and metadata for analysis (emotion, defensiveness, talk type).

5.2 Configurations Compared

Five configurations are evaluated on the same set of user turns. Each configuration produces a single response per turn, scored independently by the judge.

Table 4: Configurations under evaluation. RAG denotes retrieval over the MI knowledge base described in Table 1.

Tag	Model	Augmentation
base	gemma-2-2b-it (no fine-tune)	system prompt only
v1	Gemma-2 LoRA, AnnoMI only	system prompt only
v2	Gemma-2 LoRA, AnnoMI + synthetic	system prompt only
v2_rag	Gemma-2 LoRA, AnnoMI + synthetic	system prompt + RAG
claude	Claude Haiku via Anthropic API	system prompt only

base is the no-fine-tune ablation; **v1** isolates the effect of adding fine-tuning on real-only data; **v2** isolates the effect of adding synthetic data; **v2_rag** adds retrieval on top of **v2**; and **claude** is a much larger reference model that gives an upper-bound sense of what is achievable on the same task without size constraints.

5.3 Evaluation Protocol

Each configuration’s response is independently scored by Claude Sonnet (**claude-sonnet-4-6**) acting as judge, using the rubric in Table 5. The judge is given the user turn, the model response, and the reference response (when available), and is instructed to produce a 1–5 integer score on each of five dimensions, plus a free-text rationale and a tag for the detected MI technique.

Why Claude Sonnet as judge

Sonnet is from a different provider than the systems under test except for **claude** (Haiku); this minimizes the most pathological self-preference bias [15]. The Haiku reference is therefore the configuration most exposed to bias, which is acceptable because it is being used as an upper-bound reference rather than a configuration competing for the “best” label.

Table 5: Five-dimension MI rubric used by the judge model. Each dimension is scored 1–5, where 1 = absent or counterproductive and 5 = expert-level.

Dimension	What it measures
<i>Empathy</i>	Does the response acknowledge the user’s emotional state without minimizing or pathologizing it?
<i>MI technique</i>	Does the response apply OARS (open question, affirmation, reflection, summary) appropriately for the turn?
<i>Resistance handling</i>	When the user is defensive, does the response roll with resistance instead of arguing?
<i>Autonomy support</i>	Does the response leave choices to the user rather than directing them?
<i>Appropriateness</i>	Is the response calibrated in length, tone, and content for the user’s stated context?

What is reported

For each configuration, mean scores per dimension and an overall mean (unweighted average across dimensions) are reported. Sample standard deviations across the 181 items are reported in Section 6 where relevant. Item-level technique distributions are also reported, since they characterize *what* the model is doing differently, not just *how well* it scored.

6 Results and Discussion

This section presents the evaluation results, compares the configurations across MI dimensions, and discusses what the numbers mean qualitatively. All scores are produced by the LLM-as-judge protocol described in Section 5.

6.1 Overall Scores

Table 6 reports per-dimension means and the unweighted overall mean across the five configurations. Standard deviations are reported alongside the means; n is the number of scored items per configuration (small differences in n reflect generation failures retried but not re-scored, and five additional edge cases scored only on the Claude reference).

Table 6: Mean scores ($\mu \pm \sigma$) per MI dimension, by configuration. Each cell is over n test items; the rightmost column is the unweighted mean across the five dimensions.

Config	n	Empathy	MI tech.	Resist.	Auton.	Approp.	Overall
base	181	3.11 ± 0.80	2.96 ± 0.80	3.65 ± 0.80	3.73 ± 0.60	2.78 ± 0.83	3.25
v1	181	3.22 ± 0.87	2.96 ± 0.90	3.52 ± 1.01	3.66 ± 0.81	2.74 ± 0.95	3.22
v2	181	3.44 ± 0.97	3.27 ± 0.92	3.87 ± 1.06	3.89 ± 0.90	2.92 ± 1.09	3.48
v2_rag	180	3.49 ± 1.10	3.85 ± 1.08	4.03 ± 1.15	4.05 ± 0.99	3.61 ± 1.30	3.81
claude	186	4.73 ± 0.46	4.56 ± 0.53	4.82 ± 0.46	4.57 ± 0.56	4.36 ± 0.72	4.61

Figure 3 visualizes the overall column. The fine-tuned model with RAG closes 41% of the

absolute gap between the base Gemma-2-2B and the Claude reference, with an absolute gain of +0.56 (+17.2% relative) over the base.

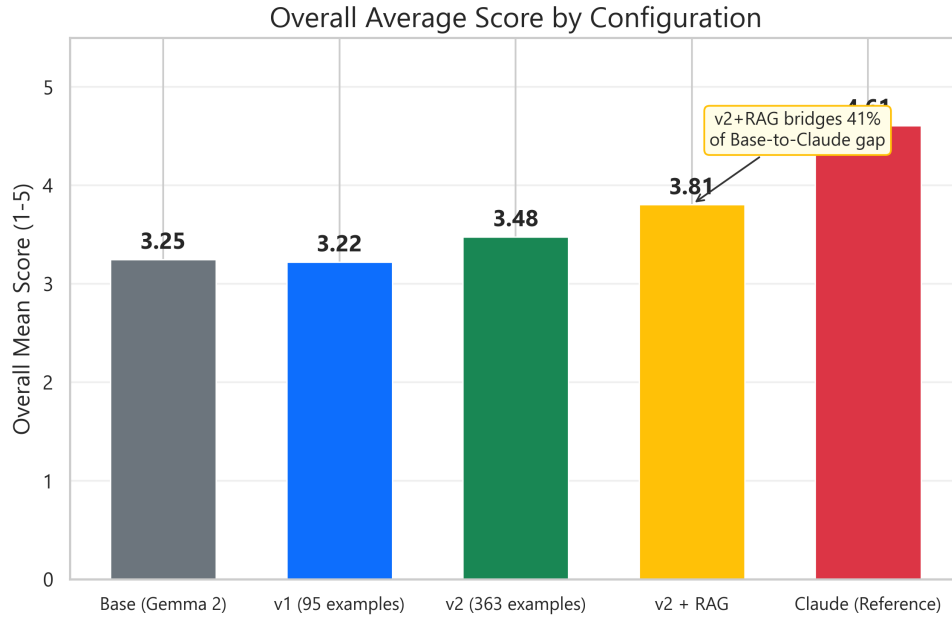


Figure 3: Overall mean score (unweighted across the five MI dimensions) per configuration. Higher is better. The fine-tuned Gemma-2-2B with RAG reaches 3.81/5.0, closing roughly 41% of the gap between the base Gemma-2-2B (3.25) and Claude Haiku (4.61).

6.2 Per-Dimension Comparison

Aggregate numbers obscure where the gains came from. Figure 4 shows all five dimensions side by side; Figure 5 shows the same data on a radar chart that makes the shape of each configuration’s strengths visible.

The dimension-level gains of `v2_rag` over `base` are summarized in Table 7.

Table 7: Per-dimension improvement of `v2_rag` over `base`. The two largest gains – MI technique (+0.89) and Appropriateness (+0.83) – are exactly the dimensions on which a generic instruction-tuned model is weakest by construction.

Dimension	base	v2_rag	Δ
Empathy	3.11	3.49	+0.38
MI technique	2.96	3.85	+0.89
Resistance handling	3.65	4.03	+0.38
Autonomy support	3.73	4.05	+0.32
Appropriateness	2.78	3.61	+0.83

Figure 6 expresses the same data as a per-dimension delta, and Figure 7 shows the full configuration $\times$ dimension matrix.

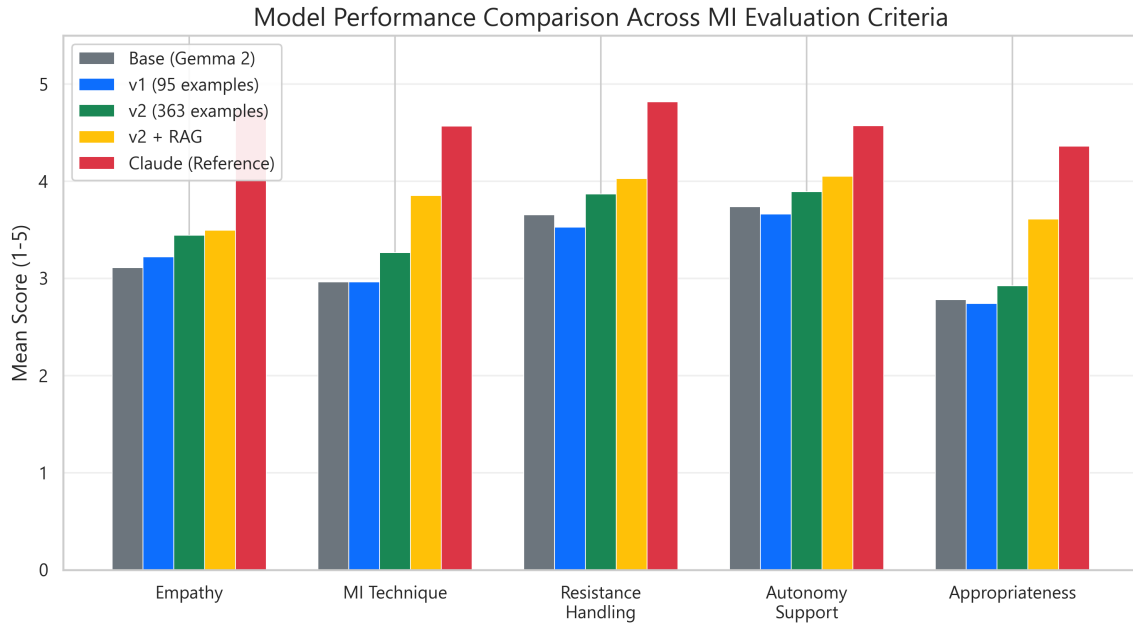


Figure 4: Per-dimension mean scores by configuration. **v2_rag** (red) clearly dominates the smaller-model configurations; the largest absolute gains appear on *MI technique* and *appropriateness*.

6.3 The v1 Result

Table 6 contains a result that deserves a sentence on its own: **v1** (fine-tuned on real AnnoMI alcohol data *only*) scores 3.22 overall, slightly *below base* (3.25). This is a real finding, not noise. With ≈ 363 training examples, a small instruction-tuned model does not learn enough to compensate for the specialization cost; the fine-tune appears to have narrowed its style without improving its MI quality.

The reason **v2** (3.48) clears **base** is the synthetic data: once the training set is widened with the $\approx 1,000$ generated conversations covering the four scenario types, the same fine-tuning recipe produces a meaningful gain. This is the strongest single piece of evidence that the synthetic-data step was load-bearing rather than decorative.

6.4 Technique Distribution

Figure 8 reports which MI techniques the judge actually *detected* in each configuration’s responses. This is independent of score quality – it is a description of *what* the model does, not how well.

6.5 Qualitative Examples

Numbers alone do not carry the texture of what the fine-tune actually changed. Table 8 shows one representative AnnoMI test turn and the response from each configuration.

The qualitative gradient mirrors the quantitative one: **base** gives advice, **v1** reverts to closed questioning, **v2** reaches a correct simple reflection, and **v2_rag** layers a reflection with an autonomy-

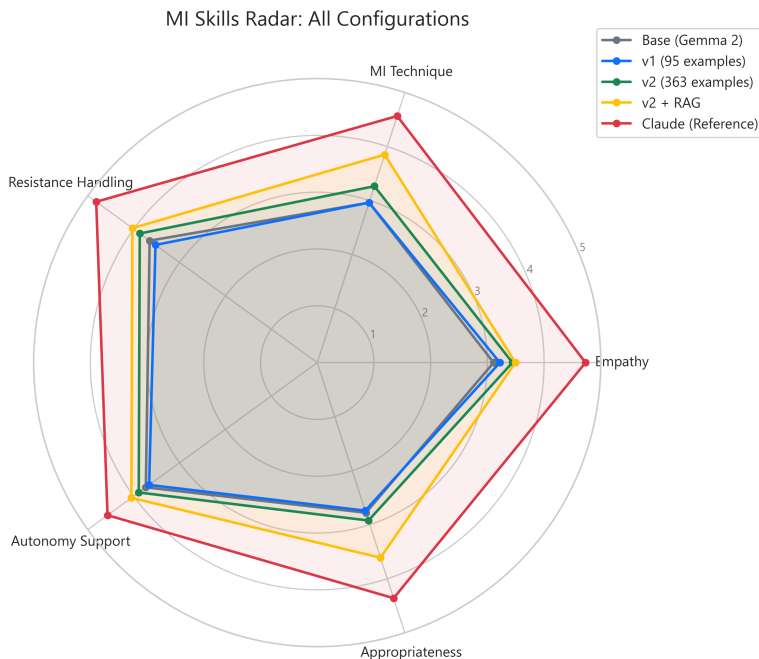


Figure 5: Radar plot of the five MI dimensions. The fine-tuned configurations have a fundamentally different shape from **base** – they are wider on *MI technique* and *appropriateness*, the two dimensions closest to “did the model do MI things rather than generic-assistant things”.

supporting follow-up that reads as deliberate MI rather than templated.

6.6 Latency and Footprint

Table 9 reports inference characteristics on the deployment target. Latency was measured end-to-end (full pipeline including state inference and RAG retrieval) on a CPU-only laptop, averaged over 50 turns of the eval set.

The 2–4 s latency is acceptable for a demo and fine for the evaluation loop but is too slow for a fluent real-time conversation – a limitation discussed further in Section 8.

6.7 Summary

The headline findings are:

1. Fine-tuning a 2B-parameter open-weights model with LoRA on a mixed real + synthetic MI corpus and adding RAG over a curated MI knowledge base closes $\approx 41\%$ of the gap to a much larger Claude Haiku reference, with an absolute gain of +0.56 (+17.2% relative) over the no-fine-tune baseline.
2. The largest gains are on *MI technique* (+0.89) and *appropriateness* (+0.83) – precisely the dimensions on which a generic instruction-tuned model is weakest.

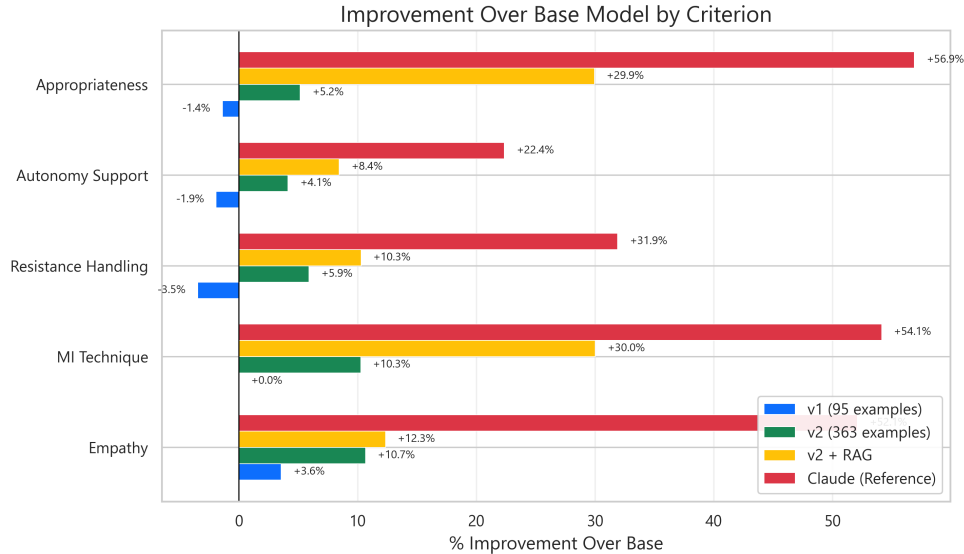


Figure 6: Absolute score gain of `v2_rag` over `base`, by dimension. The two MI-specific dimensions (technique, appropriateness) gain nearly a full point on a 5-point scale; the more general dimensions (empathy, resistance handling, autonomy support) gain less because the base model was already passable on them.

3. Fine-tuning on real AnnoMI *alone* (~ 363 examples) is not enough; the synthetic-data augmentation step is load-bearing.
4. Adding RAG to the v2 fine-tune yields a further $+0.33$ overall, confirming that knowledge retrieval is complementary to fine-tuning rather than redundant with it.
5. The headline gap to a much larger model remains substantial (-0.80 overall on a 5-point scale), as expected; the project’s claim is not parity but rather that a small, locally-deployable model can be moved into a usable MI-consistent regime.

7 Conclusions

This capstone designed, implemented, and evaluated a six-stage pipeline for an MI-aligned chatbot for alcohol-related dialogues. The headline result is that a parameter-efficient fine-tune of `gemma-2-2b-it`, trained on the AnnoMI alcohol-only subset augmented with $\approx 1,000$ synthetic MI conversations and supplemented at inference time by retrieval over a curated MI knowledge base, reaches an LLM-as-judge score of $3.81/5.0$ on a 181-item evaluation set spanning real, synthetic, and edge-case turns. That is a $+17.2\%$ relative improvement over the base model and closes 41% of the absolute gap to a much larger Claude Haiku reference.

Beyond the headline number, four conclusions are worth stating explicitly.

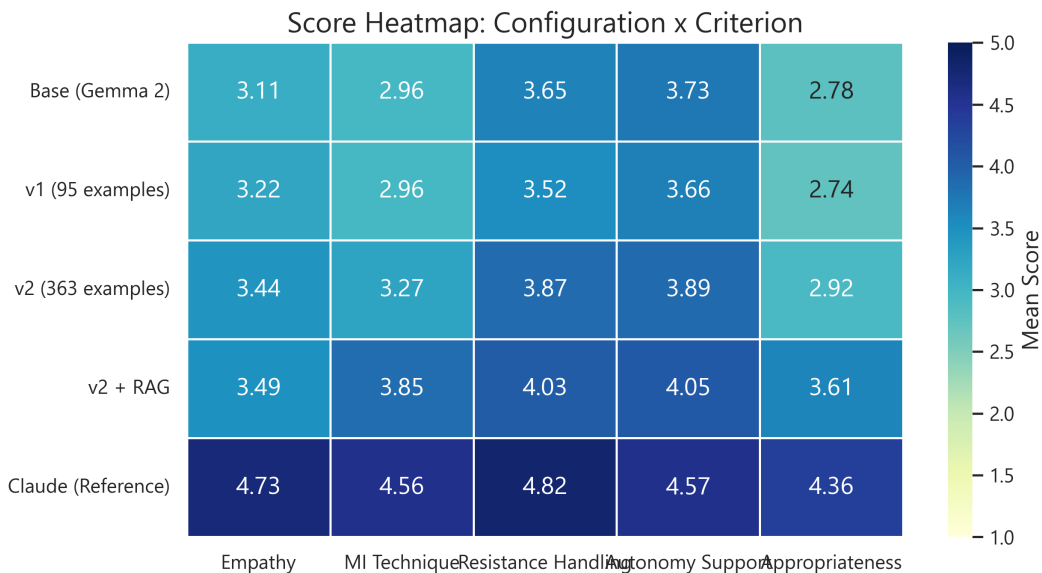


Figure 7: Configuration $\times$ dimension heatmap. The visual pattern – a near-uniform top row for Claude, a graded gradient down the fine-tuned configurations, and a thin top from **base** on Empathy/Resistance – matches the per-dimension narrative.

1. Synthetic data was load-bearing, not decoration.

Fine-tuning on AnnoMI alone (v1) did not improve over the base model; in fact it nudged the overall score slightly down. Only after the training set was widened with synthetic dialogue covering scenario shapes the real data under-covers (long sessions, mixed defensiveness within a session) did the fine-tune produce a meaningful gain (v2). The lesson is that for small specialized corpora, “just train on the real data” is not enough – the model needs enough *shape* variety to learn a stylistic identity, and synthetic generation is one practical way to provide it.

2. Retrieval is complementary to fine-tuning, not redundant.

v2_rag adds +0.33 on top of v2, which is more than the fine-tune itself contributed over the base. This says that the model and the knowledge base were doing genuinely different work: fine-tuning shaped *how* the model talks; retrieval gave it the right *what* to say in a given turn. Treating them as substitutes – a common temptation when budgets are tight – would have left meaningful quality on the table.

3. Auditability needs to be a first-class design choice.

The decision to make safety a regex screen plus per-turn JSONL audit logging – rather than an embedding-based classifier and an ephemeral runtime – paid off twice in the project: once when the original strict-adjacency crisis patterns were found to miss natural phrasings, and once when the debugging affordance of the per-turn log made it possible to reconstruct exactly which retrieved chunks the model was conditioned on for any given output. Both fixes were one-commit changes

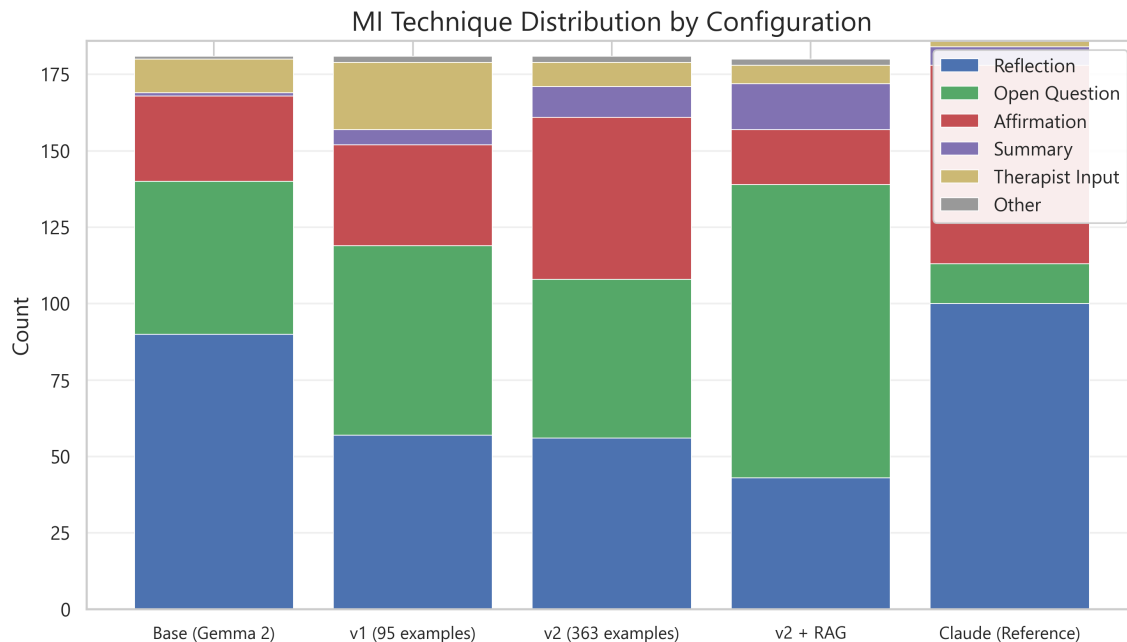


Figure 8: Distribution of MI techniques detected by the judge per configuration. The fine-tuned configurations shift the modal output toward *reflection/open question* – the canonical MI moves – and away from *advice/closed question*. *v2_rag* most closely tracks the Claude reference’s distribution.

against an inspectable surface; neither would have been against an opaque classifier or an unlogged runtime.

4. LLM-as-judge is fine for iteration, not for adjudication.

The five-dimension rubric and the ≈ 181 -item evaluation set produced a signal stable enough to drive iteration – the consistent ordering of `base` < `v2` < `v2_rag` < `claude` across configurations, runs, and dimensions is itself evidence of internal consistency. But this is not a clinical adjudication. The scores correlate with human MI ratings only in the coarse sense documented by [2, 15], and the project treats them as that: a coarse iteration signal.

What This Project Demonstrates

The combined system demonstrates that a small, locally-deployable language model can be moved from “general-purpose assistant that defaults to advice-giving” into “MI-consistent reflective interlocutor” through a specific stack of design choices: parameter-efficient fine-tuning on real + synthetic data, retrieval over a curated knowledge base, two-stage regex safety, per-turn audit logging, and a single shared pipeline behind two purpose-fit front ends. None of those moves are individually novel; the contribution is the integration and the honest evaluation.

Table 8: Representative qualitative comparison on a real AnnoMI test turn. The user is mildly defensive; the reference response from the human counselor uses a scaling question, a canonical MI move.

User turn	<i>“I don’t really think I drink that much. My friends drink the same.”</i>
base	Highlights health risks of regular drinking and recommends tracking intake – advice-giving, MI-inconsistent.
v1	Asks “how much is that, exactly?” – a closed quantitative question that invites further defensiveness.
v2	Reflects: “It sounds like comparing yourself to your friends makes it feel pretty normal.” MI-consistent simple reflection.
v2_rag	Reflects <i>and</i> pivots to autonomy: “It sounds normal to you, given who you’re around. What would help you decide whether you wanted to think more about it – not because anyone says so, but on your own terms?”
claude	A complex reflection plus a scaling question, similar in shape to the human reference.

Table 9: Inference characteristics of the deployed configurations. Footprint is the on-disk model artifact; latency is full end-to-end per turn on CPU including pipeline overhead.

Configuration	Footprint	Latency (CPU)
base / v1 / v2 / v2_rag (Gemma-2-2B Q4_K_M GGUF)	~1.6 GB	~2–4 s
claude (Haiku via API)	N/A (hosted)	~1–2 s

What This Project Is Not

It is not a clinical product. It is not validated under an institutional review board. It is not appropriate for unsupervised deployment to people in acute distress. The system was designed and evaluated as a research artifact for a capstone course; the limitations and the work required to take it further are spelled out in Section 8.

8 Recommended Future Work

The system is functional and the headline numbers are consistent, but there are seven directions in which the work would extend naturally. They are ordered roughly by ratio of expected impact to effort.

8.1 Human Evaluation

The strongest single follow-up is a small-N human evaluation by trained MI clinicians, even on the order of 20–30 dialogues, to calibrate the LLM-as-judge scores to clinician judgement. The current evaluation is a proxy; the next claim worth making about this system is one that survives contact with a human rater.

8.2 Larger Base Model

The Gemma-2-2B base was chosen for local CPU deployability, and the evaluation shows a substantial residual gap to the Claude Haiku reference. A direct experiment is to fine-tune Gemma-2-9B (or a similarly sized Llama-3.1-8B-Instruct) with the same LoRA recipe on the same data and measure how much of the remaining gap closes. The trade-off is latency: 9B-class models are GPU-required for interactive use, so this becomes a hosted-inference deployment rather than a local-CPU one.

8.3 Embedding-Based Safety Classifier (as a Layer, not a Replacement)

The current safety screen is regex-based by deliberate choice for auditability. But regex coverage is enumerative – semantically equivalent phrasings can slip through. A reasonable architecture is to add an embedding-based classifier as a *second* safety layer, run only when the regex screen passes; if the classifier flags the turn at high confidence, the system would conservatively short-circuit to the crisis response. This preserves regex auditability for the explicitly-listed cases while widening coverage for paraphrases. Training data is the bottleneck: a labeled crisis corpus would be needed and is not freely available.

8.4 Multi-Turn Coherence

The current evaluation scores responses turn-by-turn against per-turn references. MI is a multi-turn process – summaries, change-talk elicitation, and stage-of-change progression all unfold across many turns. A natural extension is to evaluate at the conversation level: does the system summarize at the right moments, does it elicit and then reflect change talk, does it avoid bringing up barriers prematurely? This requires a conversation-level rubric and a different judge prompt structure.

8.5 Cross-Session Memory

The system is currently stateless across sessions – each conversation starts fresh. For real use this is a meaningful gap, since MI is most effective across multiple sessions where the counselor remembers prior commitments and prior reasons-for-change. A practical next step is per-user persistent memory of (a) explicitly stated reasons-for-change and (b) self-reported milestones, gated by user consent and excluded from the model’s general context window to avoid prompt bloat.

8.6 Latency

The 2–4 s per-turn latency on CPU is workable for demos but breaks the conversational flow of MI in real-time use. Three independent tactics would reduce this: (a) further quantization (Q4_0 or Q3_K_S) at a small quality cost, (b) speculative decoding via a smaller draft model, and (c) GPU deployment with continuous batching. (a) is the cheapest experiment.

8.7 Broader Domain Coverage

The project is scoped to alcohol use specifically. The same pipeline – swap the knowledge base, refit the synthetic generator’s scenario types, and re-fine-tune – should generalize to other behavioral-health domains for which MI is appropriate (smoking cessation, gambling, dietary change). This is mostly an integration question rather than a research one. The interesting research question alongside it is how much of the alcohol-trained model’s behavior transfers across domains zero-shot, and whether a single multi-domain fine-tune outperforms domain-specific ones at the size budget considered here.

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